TELECOM CHURN ANALYSIS

1. Project Overview and Objective

- This project analyses customer churn for a California-based telecommunications company that provides phone and internet services.
- The objective is to identify patterns and factors associated with customer attrition by examining customer demographics, service subscriptions, contract types, tenure, and billing information. Using Excel for data cleaning and exploratory analysis and Power BI for interactive visualization, the project aims to generate actionable insights that support data-driven customer retention strategies and help reduce customer churn.

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2. Data Sources

Source Description and Timeline: https://mavenanalytics.io/data-playground/telecom-customer-churn

Domain: Telecommunication

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3. Problem Statement

- Does the churn revenue is higher than stayed and joined customer revenue. If so, then analyse their contr act type?
- Analyse the churn reason to find out the exact cause of why the customer is leaving the service
- The Internet or phone service provided by the company is causing the customer to churn?
- Is the additional charges get by the service provider causing the customers to leave?
- Does the offers provided by the telecom service has any impact?
- Does Gender and Marital status has anything to do with churn reason?

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4. Attribute (Column /Features) Details:

Table	Field	Data Type	Description
Fact_Data	Customer ID	String/Text	Unique identifier of each customer
	Tenure in Months	Numeric/Number	Indicates the total amount of months that the customer has been using the telecom service
	Avg Monthly Long Distance Charges	Numeric/ Currency	Indicates the customer's average long distance charges, (if the customer is not subscribed to phone service, this will be 0)
	Avg Monthly GB Download	Numeric/Number	Indicates the customer's average download volume in gigabytes, (if the customer is not subscribed to internet service, this will be 0)
	Unlimited Data	Categorical/Text	Indicates if the customer has paid an additional monthly fee to have unlimited data downloads/uploads: Yes, No (if the customer is not subscribed to internet service, this will be No)
	Contract	Categorical/Text	Indicates the customer's contract type
	Payment Method	Categorical/Text	Indicates how the customer pays their bill
	Monthly Charge	Numeric/ Currency	Indicates the customer's total monthly charge for all their services used from the company
	Total Charges	Numeric/ Currency	Indicates the customer's total charges for all the services used
	Total Refunds	Numeric/ Currency	Indicates the customer's total refunds
	Total Extra Data Charges	Numeric/ Currency	Indicates the customer's total charges for extra data downloads

	Total Long Distance Charges	Numeric/ Currency	Indicates the customer's total charges for long distance above those specified in their plan
	Total Revenue	Numeric/ Currency	Indicates the company's total revenue from this customer, (Total Charges + Total Extra Data Charges + Total Lond Distance Charges - Total Refurnds)
	Customer Status	Categorical/Text	Tells the customer is Churned, Stayed, or Joined
	Number of Referrals	Numeric/Number	Indicates the number of times the customer has referred a friend or family member
Customer Data	Gender	Categorical/Text	Tells the Gender of the customer
	Age	Integer/Text	Current Age of the Customer
	Married	Categorical/Text	Indicates Marital status
	Number of Dependents	Numeric/Number	Indicates the number of dependents that live with the customer (dependents could be children, parents, grandparents, etc.)
	City	String/Text	The city of the customer's primary residence in California
	Zip Code	Numeric/Text	The zip code of the customer's primary residence
	Latitude	Numeric/Text	The latitude of the customer's primary residence
	Longitude	Numeric/Text	The longitude of the customer's primary residence
Service Data	Phone Service	Categorical/Text	Indicates if the customer subscribes to home phone service
	Multiple Lines	Categorical/Text	Indicates if the customer subscribes to multiple telephone lines
	Internet Service	Categorical/Text	Indicates if the customer subscribes to Internet service
	Internet Type	Categorical/Text	Indicates the customer's type of internet connection: DSL, Fiber Optic, Cable
	Online Security	Categorical/Text	Indicates if the customer subscribes to an additional online security service provided by the company
	Online Backup	Categorical/Text	Indicates if the customer subscribes to an additional online backup service provided by the company
	Device Protection Plan	Categorical/Text	Indicates if the customer subscribes to an additional device protection plan
	Premium Tech Support	Categorical/Text	Indicates if the customer subscribes to an additional technical support
	Streaming TV	Categorical/Text	Indicates if the customer uses their Internet service to stream television programing from a third party provider at no additional fee
	Streaming Movies	Categorical/Text	Indicates if the customer uses their Internet service to stream movies from a third party provider at no additional fee
	Streaming Music	Categorical/Text	Indicates if the customer uses their Internet service to stream music from a third party provider at no additional fee
Contract Data	Offer	Categorical/Text	Identifies the last marketing offer that the customer accepted
	Paperless Billing	Categorical/Text	Indicates if the customer has chosen paperless billing
	Payment Method	Categorical/Text	Indicates how the customer pays their bill
	Tenure In Years	Categorical/ Number	Tells the no of years customer is using the telecom service
Churn Data	Churn Category	Categorical/Text	A high-level category for the customer's reason for churning, which is asked when they leave the company

	Churn Reason	Categorical/Text	A customer's specific reason for leaving the company, which is asked when they leave the company

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5. Tools & Technologies

Excel:

Data cleaning and Transformation.

Handling Missing values

Created pivot table for initial analysis

Using Star schema created fact and dimension table

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6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

Data Cleaning & Transformation:

- Using Trim, Proper and clean for text column.
- Handled missing value for AVERAGE MONTHLY LONG DISTANCE CHARGE.

  Formula used → blank value is replaced if Phone Service is "No"

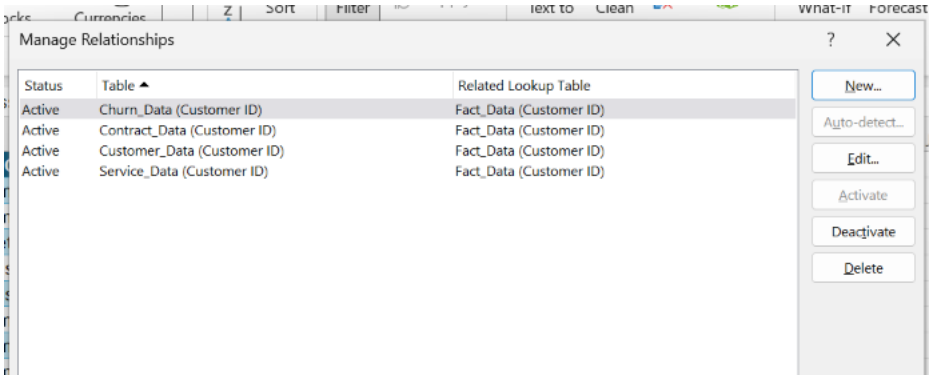
  =IF([@[Phone Service]]="No",0,[@[Avg Monthly Long Distance Charges]])
- Blank value is replaced for columns MULTIPLE LINES, INTERNET TYPE, AVERGAE MONTHLY GB DOWNLOAD, ONLINE SECURITY, ONLINE BACKUP, DEVICE PROTECTION PLAN, CHURN CATEGORY, CHURN REASON
- Created a new Column TENURE IN YEARS

  Tenure in Years under Contract_Data sheet

  =IF([@[Tenure in Months]]<=12,"1 Year",IF(AND([@[Tenure in Months]]>12,[@[Tenure in Months]]<=24),"2 Years",IF(AND([@[Tenure in Months]]>24,[@[Tenure in Months]]<=36),"3 years",IF(AND([@[Tenure in Months]]>36,[@[Tenure in Months]]<=48),"4 Years",IF(AND([@[Tenure in Months]]>48,[@[Tenure in Months]]<=60),"5 Years","6 Years")))))

Data Modeling:

Created relationship between all the tables



Pivot Tables:

Created Pivot Tables for data summarisation and initial insights.

Status	No of Customers	Total Revenue
Churned	1869	3684459.82
Joined	454	54279.75
Stayed	4720	17632392.12
Grand Total	**7043**	**21371131.69**

Contract	No of Customers	Sum of Number of Referrals
Month-to-Month	**3610**	**4112**
Churned	1655	790
Joined	408	352
Stayed	1547	2970
One Year	**1550**	**3687**
Churned	166	140
Joined	24	44
Stayed	1360	3503
Two Year	**1883**	**5948**
Churned	48	44
Joined	22	35
Stayed	1813	5869
Grand Total	**7043**	**13747**

Status	Sum of Number of Referrals
Churned	974
Joined	431
Stayed	12342
Grand Total	**13747**

Count of Customer ID	Customer Status			
Tenure in Years	Churned	Joined	Stayed	Grand Total
0	784	454	232	1470
1	406		761	1167
2	235		769	1004
3	155		560	715
4	144		724	868
5	82		704	786
6	63		970	1033
Grand Total	**1869**	**454**	**4720**	**7043**

Churn Category and Reason	No of Customers
Attitude	**314**
Competitor	**841**
Customer Stayed	**5174**
Dissatisfaction	**321**
Other	**182**
Price	**211**
Grand Total	**7043**

Count of Customer ID	Unlimited Data			
Customer Status	No	Not Applicable	Yes	Grand Total
Churned	254	113	1502	1869
Joined	48	182	224	454
Stayed	470	1231	3019	4720
Grand Total	**772**	**1526**	**4745**	**7043**

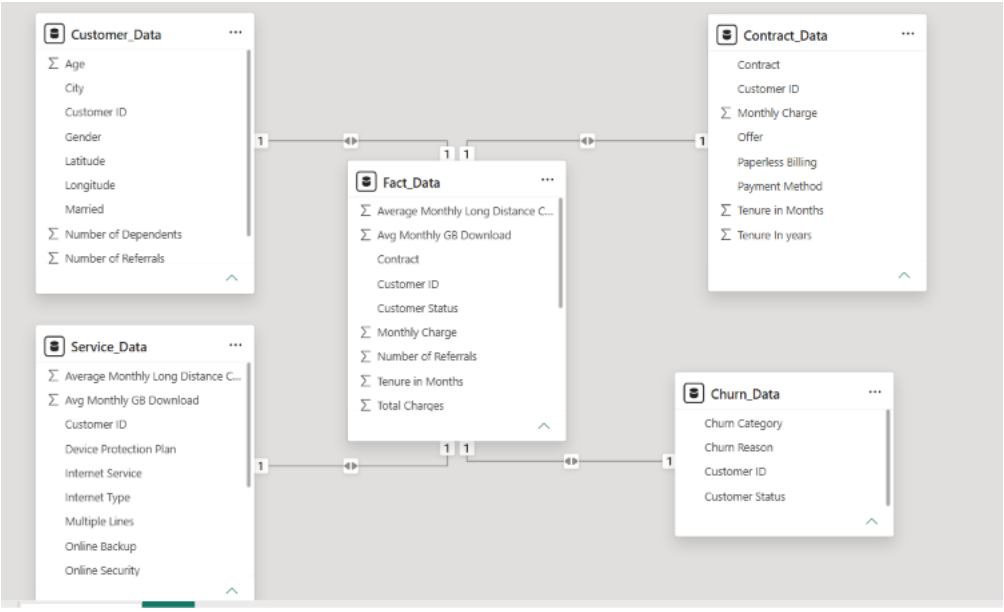
Count of Customer ID	Customer Status			
Internet Service	Churned	Joined	Stayed	Grand Total
No	113	182	1231	1526
Yes	1756	272	3489	5517
Grand Total	**1869**	**454**	**4720**	**7043**

Customer Status	Sum of Total Long Distance Charges
Churned	810991.9
Joined	17309.2
Stayed	4447605
Grand Total	**5275906.1**

- Converted data into Fact and Dimension Table

7. Data Modelling and DAX (Power BI)

- **Data Model: Imported 5 tables from excel – Fact_Data, Customer_Data, Service_Data, Contract_Data, Churn_Data and connected using Customer ID as this is the only common primary key across all tables.**



- **DAX Measures:**

  Dax measures used to create important KPIs

```
1. Total Customers = COUNT(Fact_Data[Customer ID])
2. Customer Stayed = CALCULATE(COUNT(Fact_Data[Customer ID]),Fact_Data[Customer
   Status]="Stayed")
3. Churned Customer = CALCULATE(COUNT(Fact_Data[Customer
   ID]),Fact_Data[Customer Status]="Churned")
4. Customer Joined = CALCULATE(COUNT(Fact_Data[Customer ID]),Fact_Data[Customer
   Status]="Joined")
5. Overall Revenue = SUM(Fact_Data[Total Revenue])
6. Revenue_CustomerStayed = CALCULATE(SUM(Fact_Data[Total
   Revenue]),Fact_Data[Customer Status]="Stayed")
7. Revenue_CustomerChurned = CALCULATE(SUM(Fact_Data[Total
   Revenue]),Fact_Data[Customer Status]="Churned")
8. Revenue_CustomerJoined = CALCULATE(SUM(Fact_Data[Total
   Revenue]),Fact_Data[Customer Status]="joined")
```

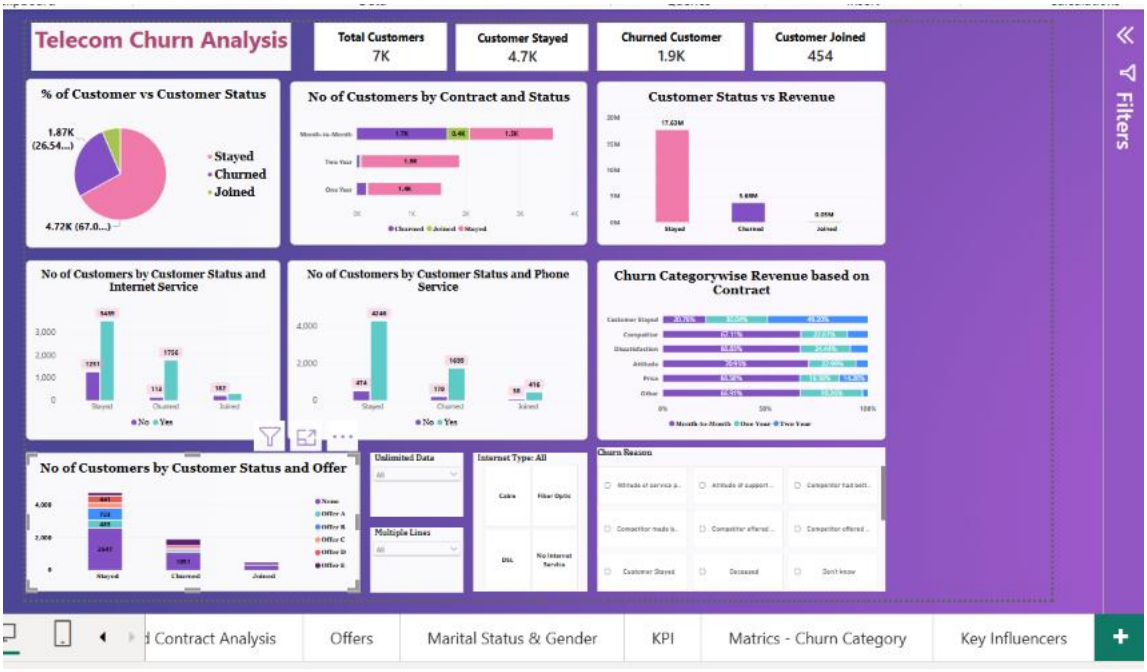
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8. Analysis and Visualizations (Power BI)

Dashboard Features:

- **Pie chart shows the % of customers status through which you came to know the reason for this analysis.**
- **Revenue stats is represented in bar chart.**
- **Imapct of Phone and Internet service and Contract based impact is represented in column and stacked bar chart.**
- **Offer based impact is shown in stacked column chart**

- **Slice includes churn reason, internet type, unlimited data and multiple lines which helps to analyse deeper in the dashboard view.**
- **Further analysis is represented in different pages using different charts and matrics tables and key influencers.**



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9. Insights & Conclusions

Key Findings & Insights:

Descriptive:

- Analysed the customer behaviour based on contract type, phone and internet services used by the customer.
- Analysed churn reasons and category to make better decisions in the future
- Analysed some usage patterns of customer like how many users opt for the offers provided by the telecom service.
- Analysed the data based on gender and marital stauts

Diagnosis

- 26% of customer is churned which is almost half the value of stayed customer. Newly joined customer is only 6.45% which is alarming and indicates that we need to improve our service.
- Compared to other to contract types month-to-month contract attracts more customers but generates less revenue because the churn % is high in month-to-month contract.
- Most of the churned customer says reason is “Competitor” which indicates we have better competitor.
- Other churn reasons are dissatisfaction and attitude which indicates support system is poor.
- Offer E customer is likely to churn more rather than customer who uses other offers or doesn’t use any offer.
- No of customer churn is less those who use some offer.
- Unmarried customer(1200) are likely to churn more than married customer(669).

Predictive

- Long term customer will likely to move if we don't improve our service.
- More new customers will likely to join on Month-to-Month contract.

Prescriptive

- Customer will likely to move to other service which is alarming. hence improvement in all the service(Phone and Internet) is mandatory.
- Providing more offers in month-to-month contract will hold the customer
- Support persons need to be monitored or they should be trained to handle customer smoothly. As most of the customer leave the service is because of support persons behaviour.
- Apart from price we need to improve customer service, product enhancement and promoting offers.
- Providing some offers for unmarried customer will be a better choice.

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10. Conclusions

This project analysed the customer behaviour to retain the customer and to decrease the churn rate. Using Excel for Data cleaning and transformation and Power BI to find out the reason for leaving using different charts and matrics tables to find out patterns and insights.